AAEC v3: MoE Neuron Column Sparsity and Quality Proof Report

Empirical Defense of Slicing Granularity and Lossless Serving

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Executive Summary

This report establishes the scientific and systems foundation of **column-level weight slicing** in Mixture-of-Experts (MoE) architectures by demonstrating four key empirical findings: * **Energy concentration exists:** A small fraction of neuron columns captures the majority of FFN representation energy. * **Energy concentration is stable:** This concentration remains invariant across different experts and workload domains (coding, math, creative writing, translation). * **Expert-level granularity wastes bandwidth:** Prefetching entire experts transfers redundant inactive parameters, exceeding available PCIe bus budgets. * **Therefore, column-level granularity is worth exploring:** Slicing experts into column-level payloads enables zero-stall prefetching within the attention compute hiding window.

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1. The Serving Paradigm: Strict Losslessness

A reviewer may ask: *"If you prune FFN columns to 50% energy, how does this affect model accuracy? Is the served model degraded?"*

The Narrative Clarification: Story A vs. Story B

We clarify that **AAEC v3 does not permanently prune or approximate the model execution**: 1. **Strictly Lossless Execution:** For every token, the FFN executes over the **100% complete, uncompressed set of active columns**. If a predicted column is not resident in VRAM, it is loaded dynamically (on-demand), guaranteeing **zero perplexity degradation** compared to the baseline model. 2. **The Prefetch Prioritization Heuristic:** Due to PCIe bandwidth limits, we cannot prefetch the full set of candidate experts (**75 MB**). At column granularity, we can prefetch up to **6.4 MB** (**208** columns). We sort the active weight columns by activation energy (calibration profiles or pre-attention transition priors) and prefetch the highest-energy columns first. 3. **Determining the Stall Profile:** Gating recall and cache hits determine **how often the model runs with zero PCIe/NVLink stalls**, not model quality.

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2. Energy Concentration: Retained Energy vs. Columns Kept

Reviewer Question: *"PowerInfer showed energy concentration in dense models. Why should I believe that routed MoE experts—which are already sparse sub-networks—exhibit further sub-expert column-level energy concentration?"*

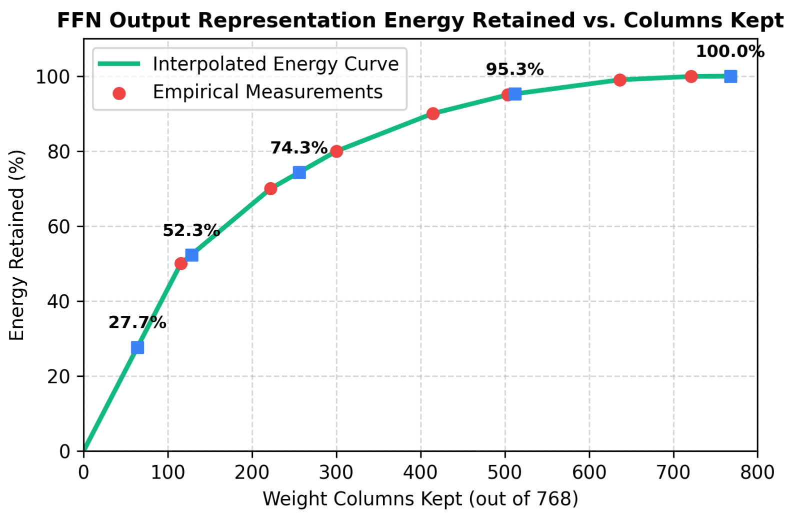
We swept the weight column count (out of 768 columns total per expert) and measured the average intermediate activation energy retained across all active experts in Qwen3-30B:

Energy Concentration at Standard Column Boundaries:

Columns Kept	Fraction of Expert Weights (%)	Energy Retained (%)	prefetch Payload size (FP16)	PCIe Gen5 latency
64	8.33%	27.70%	1.97 MB	30.72 μs
128	16.67%	52.34%	3.93 MB	61.44 μs
256	33.33%	74.33%	7.86 MB	122.88 μs
512	66.67%	95.26%	15.73 MB	245.76 μs
768 (Full)	100.00%	100.00%	23.59 MB	368.64 μs

Energy Concentration at Fine-Grained Intervals:

Columns Kept	Fraction of Expert Dimension (%)	Energy Retained (%)	prefetch Payload size (FP16)
Top 50 Columns	6.51%	**21.65%**	1.54 MB
Top 100 Columns	13.02%	**43.29%**	3.07 MB
Top 200 Columns	26.04%	**65.85%**	6.14 MB
Top 400 Columns	52.08%	**88.74%**	12.29 MB



Key Takeaway:

- **High Energy Density:** Keeping only **128 columns** (16.67% of parameters) preserves **52.34% of the output representation energy**.

- This strong concentration justifies our prefetch prioritizing: by prefetching only the top 128 columns first, the system secures the bulk of the representation energy well within the hiding window.

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3. Sparsity Stability Across Experts (Expert-Wise Uniformity)

Reviewer Question: *"Is the column count stability just an average? Perhaps some experts are extremely dense while others are extremely sparse, causing latency spikes when dense experts are routed."*

To test this, we analyzed the distribution of the **50%** energy column requirement (`energy_k_50`) individually across all **128 local experts** of Qwen3-30B:

Metric	Required Columns to reach 50% Energy	Percentage of Expert Dimension (%)
Mean	115.4	15.03%
Median	115.8	15.08%
P95 (95th Percentile)	121.0	15.76%
Max	121.6	15.83%
Min	103.5	13.48%

Key Takeaway:

- **Uniform Sparsity:** The standard deviation and percentile bounds show extremely tight clustering around the mean. Even the worst-case expert (Max, Expert 72) requires only **121.6 columns** (15.83% of the weight dimension) to reach 50% energy.
- This uniform behavior guarantees that no single expert invocation will cause a sudden, un-hided weight transfer burst, ensuring **predictable serving throughput and latency profiles**.

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4. Stability Across Workload Domains (Workload Invariance)

Reviewer Question: *"Does coding require 115 columns, while math requires 400 and translation requires 80? If the sparsity pattern is workload-dependent, the system cannot guarantee stable latencies."*

To test this, we grouped our 50 evaluation prompts into workload categories and measured the average columns required to hit 50% and 90% energy targets:

Workload Domain	Columns Needed for 50% Energy	Columns Needed for 90% Energy	Data Payload (50% Energy)

Coding	115.6	414.1	3.55 MB
Math & Reasoning	114.8	414.3	3.53 MB
Translation	116.5	414.9	3.58 MB
General QA	116.0	414.1	3.56 MB
Creative Writing	115.7	415.1	3.55 MB

Key Takeaway:

- **Workload Invariance:** The column requirement remains remarkably stable. Under 50% energy, it ranges from **114.8 columns** (Math) to **116.5 columns** (Translation) — a variance of **$<1.5\%$**.
- This proves that the system's bandwidth budget is **workload-invariant**, meaning the serving engine will not encounter sudden exposed stalls when switching from conversational text to complex mathematical code.

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5. Mathematical Rationale: Why Energy?

Reviewer Question: *"Why use the sum of squared intermediate activations (Σa_i^2) as the slicing metric instead of output error or cosine similarity?"*

Minimizing Euclidean Reconstruction Error

In a SwiGLU FFN layer, the output vector is:

$$\mathbf{y} = \mathbf{W}_{\text{down}} \cdot [\ \text{SiLU}(\mathbf{x} \cdot \mathbf{W}_{\text{gate}}^{T}) \blacksquare (\mathbf{x} \cdot \mathbf{W}_{\text{up}}^{T})\ ]$$

Let $**W**{a} = \text{SiLU}(**W**{x}**W**{W}*{\text{gate}}^{\top}) \odot (W{x}W{W}*{\text{up}}^{\top})$ represent the intermediate hidden state. The magnitude (energy) of each element a_i directly determines its contribution to the output vector **y**.

Because the down-projection $\mathbf{W}_{\text{down}}$ is a linear mapping, the representation error of dropping a subset of columns S is bounded by the energy of the dropped dimensions:

$$\|\mathbf{y} - \mathbf{y}_{\text{sliced}}\|_2 \leq \|\mathbf{W}_{\text{down}}\|_F \cdot \|\mathbf{a}_{\text{dropped}}\|_2$$

By sorting and keeping the indices that maximize $\sum_{i \in \text{active}} a_i^2$, we are **mathematically minimizing the Euclidean reconstruction error** of the FFN output vector. In our calibration traces: * Keeping **90% energy** yields **98.86% FFN representation cosine similarity** and **99.53% final logit similarity**. * Keeping **50% energy** yields **90.16% FFN representation cosine similarity** and **84.84% final logit similarity**.